

4.3 Antimicrobial Therapies

▼ What is prontosil?

- It the first example of a **sulphonamide antibiotic**

▼ Recall two other properties of prontosil.

- Bacteriostatic
- Synthetic

▼ What is prontosil used for?

- Urinary Tract Infections (UTIs)
- Respiratory Tract Infections (RTIs)
- Bacteraemia (bacteria in the blood)

▼ What are sulphonamide antibiotics?

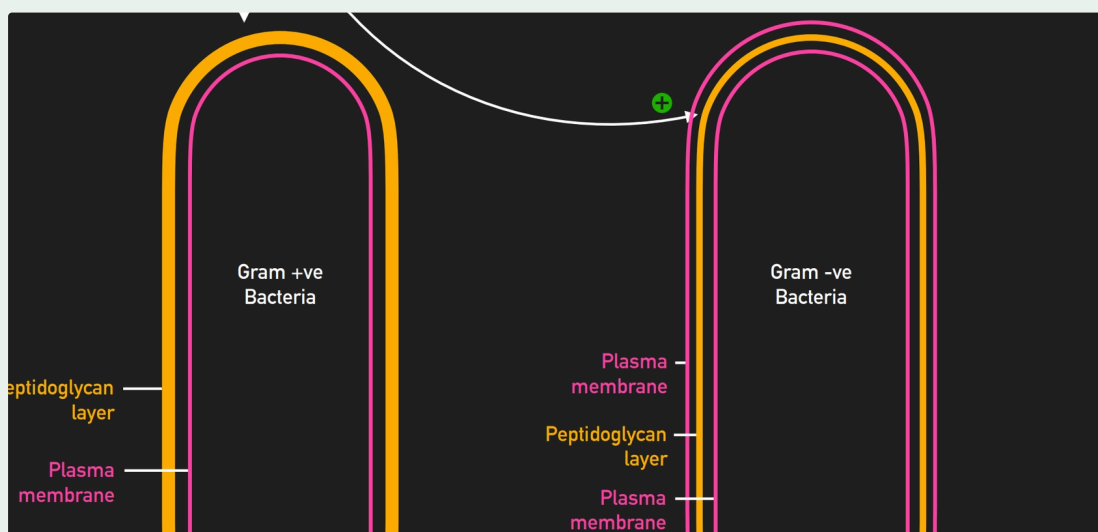
- A class of **broad-spectrum antibiotics** used to treat various types of bacterial infections.

▼ What is a drawback of prontosil?

- It is **only effective against gram-positive bacteria**.

▼ What property of gram-negative bacteria makes them more difficult to treat with antibiotics?

- The presence of an **outer and inner membrane** in gram-negative bacteria makes things difficult
- This is because the outer membrane has different properties to the inner membrane so requires different treatment to break down.



▼ What is a beta-lactam?

- The most commonly prescribed **antibiotic**

▼ Describe the mechanism through which beta-lactams function.

- They **bind to** structures on the bacterial cell wall called **penicillin binding proteins**
- This interferes with the **synthesis of the peptidoglycan component of a bacterial cell wall**

▼ Give two examples of beta-lactams.

- Penicillin
- Methicillin

▼ What are penicillin-binding proteins?

- These are proteins involved in the **synthesis of peptidoglycan** (an important component of bacterial cell walls)
- Beta-lactams bind to these, to inhibit peptidoglycan synthesis and hence **weaken bacterial cell walls**

▼ What is an antibiotic?

- An antimicrobial agent **produced by a microorganism** that **kills or inhibits other microorganisms**
- Most antibiotics used today are produced by fungi or bacteria.

▼ Describe the difference between bactericidal and bacteriostatic antibiotics.

- Bactericidal antibiotics kill bacteria
- Bacteriostatic antibiotics inhibit bacterial growth

▼ What is an antiseptic?

- An **antimicrobial substance** used to **prevent the growth of microbial agents**.

▼ What is a patient-induced cause of antibiotic resistance?

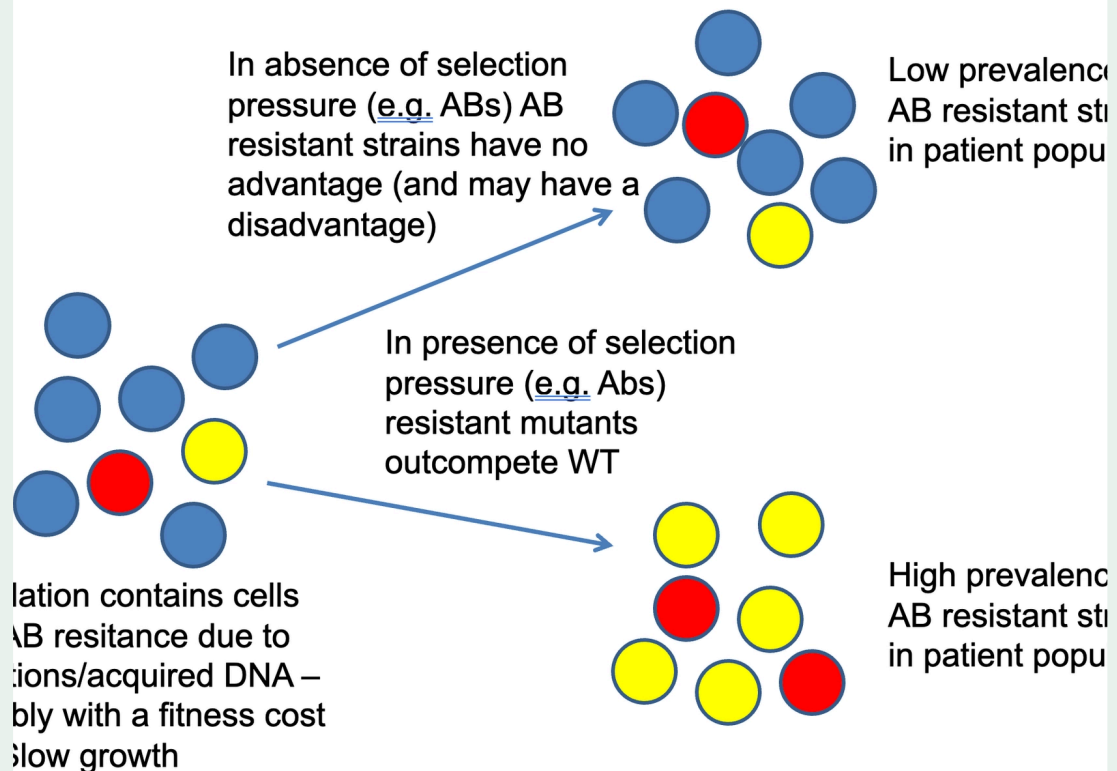
- **Underdosing**
- Using an **inadequate quantity of antibiotic** which results in the **survival of bacteria**.

▼ What is the minimal inhibitory concentration (MIC)?

- The **lowest concentration of an antibiotic** that is **required to inhibit bacterial growth**.
- You can look at it as the **maximum concentration of antibiotic** that a **bacterium can grow in**

▼ Describe the mechanism through which antibiotic resistance occurs.

Antibiotic resistance – evolution by natural selection



▼ Why is antibiotic resistance more prevalent in hospitals than in the community?

- Antibiotics are **used much more frequently in hospitals**, and so the **bacteria is subject to a much heavier selection pressure** in this environment
- This means it is **much more likely** for the **prevalence of an advantageous mutation** to increase in popularity in hospitals, than in the community as they are being selected for.

▼ What is meant by horizontal gene transfer in bacteria?

- The exchange of DNA between bacteria in the same generation.
- It is a mechanism through which bacteria can resist antibiotics.

▼ What can antibiotic resistance lead to from a clinical point of view?

- Increased mortality (death)
- Increased morbidity (illness)

- ▼ What can antibiotic resistance lead to from a financial point of view?
 - Increased cost
 - Newer drugs, more drugs may have to be used which incurs a higher financial cost.
- ▼ What is peptidoglycan made of?
 - Simple sugars and amino acids.
 - PEPTIDO - GLYCAN
- ▼ What is a bacterial cell membrane made of?
 - Protein and phospholipid.
- ▼ Recall four gram-positive bacteria and what they cause.
 - **Staphylococcus aureus** - wound/skin infections, pneumonia, septicaemia, infective endocarditis
 - **Streptococcus pneumoniae** - Pneumonia, septicaemia
 - **Clostridium difficile** - Pseudomembranous colitis, antibiotic-associated diarrhoea
 - **Enterococcus spp** - Urinary Tract Infections (UTI), gut commensal
 - **Mycobacterium tuberculosis** - Tuberculosis
- ▼ Recall four gram-negative bacteria and what they cause.
 - **E Coli** - GI infections, neonatal meningitis, urinary tract infections (UTI)
 - **Salmonella** - GI infections, typhoid fever
 - **Neisseria gonorrhoeae** - gonorrhoea
 - **Acinetobacter baumannii (MDRAB)** - Opportunistic infections, wounds, UTI, pneumonia (VAP).
 - **Pseudomonas aeruginosa** - Cystic fibrosis, burn wound infections. Survives on abiotic surfaces.
- ▼ What are aminoglycosides?
 - They are **bactericidal antibiotics**
- ▼ Describe the mechanism through which aminoglycosides function.
 - They target protein synthesis by **causing the wrong amino acids** and hence the **wrong membrane proteins to be produced**
 - This leads to the **death of bacterial cells** as damage has been done to the cell membrane

▼ What is rifampmicin?

- They are **bactericidal antibiotics**

▼ Describe the mechanism through which rifampmicin functions.

- It targets the RpoB **subunit of RNA polymerase**
- This leads to **protein synthesis not proceeding smoothly** and hence the **death of the cell**

▼ Recall a side effect of rifampmicin.

- It makes secretions go red/orange
- This leads to problems encouraging patients to comply with the full course

▼ What is vancomycin?

- It is a **bactericidal antibiotic**

▼ Describe the mechanism through which vancomycin functions.

- Targets cell wall biosynthesis

▼ What is linezolid?

- It is a **bacteriostatic antibiotic** that works against gram-positive bacteria.

▼ Describe the mechanism through which linezolid functions.

- It **inhibits the initiation of protein synthesis** by **binding** to the **50S (smaller) ribosomal subunit**

▼ What is daptomycin?

- It is a **bacteriostatic antibiotic** that works against gram-positive bacteria

▼ Describe the mechanism through which daptomycin functions.

- Targets bacterial cell membranes

▼ What is an important consideration to make when prescribing linezolid and daptomycin?

- They only work on gram-positive bacteria.

▼ What is meant when antibiotics are described as selectively toxic?

- It means that they are **selectively more toxic** (~ 100 times more) to bacterial cells than to human cells
- This is because **bacterial cells** are **significantly distinct to human cells**

▼ What are the four mechanisms through which antibiotic resistance can occur?

- Altered target site (antibiotic can no longer engage its target)
- Inactivation of antibiotic
- Altered metabolism
- Decreased drug accumulation

▼ What are the two ways bacteria can obtain altered target sites?

- Acquisition of a gene that encodes for a different target
- Acquisition of a gene that encodes for a target-modifying enzyme which makes antibiotic ineffective against target

▼ How does an altered target site lead to antibiotic resistance?

- It can lead to **antibiotics not being able to bind** onto the relevant **target sites** anymore as **their structures have changed** or the **target site has been relocated**

▼ Describe how an altered target site in MRSA leads to antibiotic resistance.

- *i.e MRSA can **encode** for an **alternative penicillin-binding protein (PBP2a)**.*
- ***Beta-lactams** can still bind onto the original PBP, but it has no effect as the **PBP2a can still help the MRSA** carry out protein synthesis.*
- *The PBP2a also has a **low affinity for beta-lactams**.*

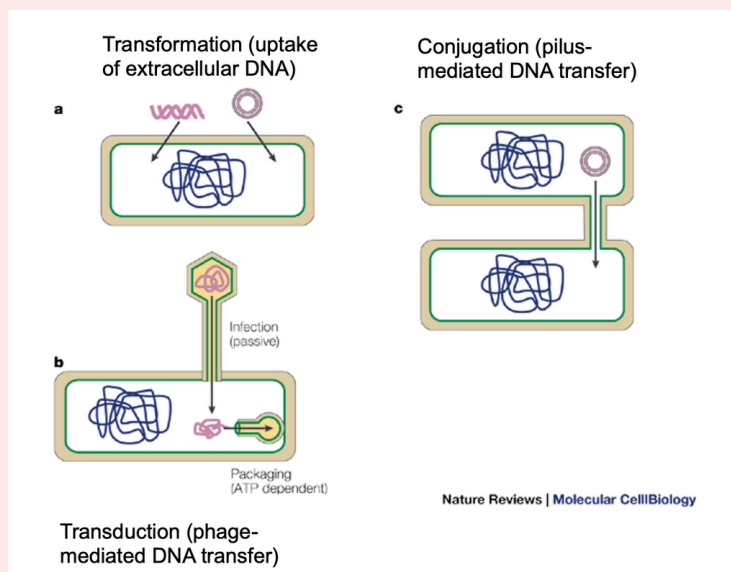
▼ How does inactivation of an antibiotic lead to antibiotic resistance?

- The **bacteria produces an enzyme** which leads to the **degradation or alteration of the antibiotic**
- This **renders it ineffective**
- An example of this is **beta-lactamase**

▼ How does altered metabolism lead to antibiotic resistance?

- Because antibiotics target central metabolic processes in bacteria (i.e protein synthesis), a method of bacterial evasion is to find other ways of obtaining energy/materials from the environment
- Bacteria can also reduce reliance on substances produced from the targeted pathways, so that any effect that antibiotics have are rendered null.

- ▼ How does decreased drug accumulation lead to antibiotic resistance?
 - This works by **pumps** on the surfaces of bacterial cell membranes **reducing the intake of antibiotic** from the environment, and **increasing the efflux of antibiotic to the environment**
 - This means the **drug does not reach the required concentration** to be effective against the bacterial cell.
- ▼ What is a macrolide and how does it work?
 - It is an **antibiotic** used to **stop polypeptide chain formation** by **binding to the 50S ribosomal subunit**
- ▼ What is a quinolone and how does it work?
 - An entirely synthetic **bactericidal antibiotic** which **inhibits DNA replication**.
- ▼ Name the three sources of antibiotic resistance genes.
 - Plasmids - these can code for multiple resistance genes
 - Transposons - Integrate DNA **from chromosomes to plasmid** and vice-versa
 - Naked DNA - **DNA released by other bacteria is taken up** from the environment
- ▼ Describe three mechanisms of antibiotic resistance gene uptake.
 - **Transformation** (uptake of extracellular DNA)
 - **Transduction** (phage-mediated DNA transfer)
 - **Conjugation** (pilus-mediated DNA transfer)



▼ What is a bacteriophage?

- A virus that affects bacteria
- It is involved in the horizontal spread of DNA throughout a bacterial population.

▼ Recall three ways that antibiotic treatment may fail.

- **Biofilm**
- **Intracellular location**
- **Slow growth**
- **Spores**
- **Persisters**

▼ Name three ways that non-genetic antibiotic resistance can arise within a population.

- Inappropriate drug choice for bacteria
- Poor penetration of antibiotic into target site
- Inappropriate dosage
- Inappropriate administration (i.e oral or intravenous)
- Presence of other bacteria which are able to kill/break down antibiotic - **collateral resistance**

▼ Name three risk factors for hospital-acquired infections.

- Large number of ill people
- Crowded wards
- Broken skin - easy entry
- Transmission by staff - poor hygiene

▼ How can we address antibiotic resistance to prevent it arising?

- Removal of selection pressures which are easy to evade (i.e broad spectrum antibiotics)

▼ How can we react to antibiotic resistance in humans?

- Modifying antibiotics to make them more effective
- **Combinations of an antibiotic and inhibitor of bacterial mechanism** (i.e PBPA2)
- The problem is, that these are responses to resistance after it has occurred.

- ▼ What are the three broad classes of condition that fungi cause in humans?
 - **Allergy** - allergic reactions to fungal products
 - **Mycotoxicoses** - ingestion of fungi and their toxic products
 - **Mycoses** - superficial, subcutaneous or systemic **colonisation, invasion and destruction of human tissue.**
- ▼ What are three targets for antifungal agents?
 - **Cell membrane** - particularly ergosterol
 - **DNA synthesis** - arrested by agents
 - **Cell wall**